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Evaluating the Effectiveness of the National Curriculum and Training Institute's "Cognitive Life Skills" Program among Parolees Supervised by Pennsylvania's Board of Probation & Parole

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INTRODUCTION

In the decades since Martinson (1974) wrote his controversial piece about "what works," many treatment professionals (e.g., counselors, therapists, program developers, academics, etc.) have learned a better and more multifaceted question to ask about prison reform is "What works for certain offenders and under what conditions?" This revised inquiry has led researchers to seek approaches to correctional treatment that generates measurable and lasting changes among offenders released from prison and returned back to society. During the past 40 years, many offender-based programs have accepted a cognitive-behavioral approach to rehabilitation as a means of combating high recidivism rates among offenders released from incarceration (see, Landenberger & Lipsey, 2005; Saxena, Messina, & Grella, 2014; Van Voorhis, Spiropoulos, Ritchie, Seabrook, & Spruance, 2013).

While cognitive-behavioral techniques have been used widely in behavioral and social disciplines to treat non-criminal and non-delinquent behaviors including insomnia (Vitiello, McCurry, & Rybarczyk, 2013), psychosis (Thomas, 2015), and obsessive compulsive disorder (Pozza, Andersson, & Dèttore, 2015), discussion here will focus on interventions used in the criminal justice system with an incarcerated population and offenders who are under community supervision. The basic notion behind cognitive-behavioral approaches to prison treatment and rehabilitation is that criminal thinking and criminal behavior are linked, and therefore changing one's criminal thoughts is the first step to changing one's criminal behavior. A cognitive-behavioral intervention (CBI) focuses on cognitive development, such as problem-solving, decision-making, and recognizing unrealistic thinking. This approach involves behavioral components, such as role playing and modeling exercises, that assess comprehension and readiness for change (Clark, 2011). Here the effectiveness of several common CBIs delivered to

offenders will be described; however, the focus of this analysis is to reveal recidivism findings from the first empirical evaluation conducted among offenders completing a CBI developed by the National Curriculum and Training Institute.

CBI Programming for Offenders

A review the relevant literature for outcomes of CBI programming for offenders yielded many results about the effectiveness of specific treatment and rehabilitative initiatives used in the correctional field. Published findings in peer-reviewed journals showed three common CBIs utilized by many correctional jurisdictions in the United States, Canada, United Kingdom, etc. to improve recidivism rates among targeted offender populations. These programs include Thinking for a Change (T4C), Moral Reconation Therapy (MRT), and Reasoning and Rehabilitation (R&R). While additional evaluation studies may exist, the review provided here identifies the major empirical works that have been published related to the effectiveness of each program. Great variation was uncovered among individual studies related to the methodologies employed and statistical rigor of the analyses performed.

Findings from these evaluation studies were mixed regarding the effectiveness of the interventions for reducing overall recidivism rates among offenders completing the programs. The T4C program attempts to develop critical problem-solving, decision-making, and social intervention skills among offenders, while identifying sources of cognitive thinking errors and criminal or anti-social beliefs, attitudes, and values (Milkman & Wanberg, 2007). Despite its popularity, only a few outcome studies were found that assess the effectiveness of the program for reducing recidivism among offenders. One empirical study that included control group offenders and program dropouts found that offenders completing the T4C program had improved

problem-solving skills, fewer technical parole violations, and a 33 percent reduction in new crimes during a 12-month follow-up period (Golden, Gatchel, & Cahill, 2006). Lowenkamp, Hubbard, Makarios, and Latessa (2009) also presented outcomes for the T4C program measured as "an arrest for a new criminal charge (misdemeanor or felony offense)" (p. 142). These findings revealed a non-trivial effect of the program, including a 28 percent adjusted recidivism rate among offenders who completed or had at least one session of T4C compared to a 43 percent adjusted recidivism rate among control group offenders not referred to the program.

Another popular CBI widely used to impact future criminal behaviors among offenders is MRT. A meta-analysis of empirical studies on the effectiveness of the MRT program on recidivism rates was conducted by Wilson, Bouffard, and MacKenzie (2005). Findings gathered from the one experimental evaluation and the three high-quality quasi-experimental studies revealed a 41 percent recidivism rate in five years among program completers compared to a rate of 56 percent in a comparison group and overall "positive effects, although half were statistically significant due to insufficient statistical power" (p.189). It was concluded that "there is reasonably strong evidence for the effectiveness of MRT at reducing long-term recidivism rates among offenders" (p. 189). However, Wilson et al., (2005) urged caution when interpreting findings because the majority of the empirical analyses were conducted by the program developers (see Little, Robinson, & Burnette, 1991; Little & Robinson, 1989).

Another more recent meta-analysis focusing on MRT was conducted by Ferguson and Wormith (2012). This analysis examined 33 evaluations of the MRT program while controlling for a number of factors, including age, gender, methodological design, sample size, location, recidivism measures, and follow-up period. While individual evaluation studies were criticized for weak research designs, program delivery failures, lack of randomized controls, and a low

mean effect size, the overall conclusion reached was that "MRT is effective in reducing recidivism" (p. 1001).

A third CBI used widely with an offender population is R&R. This intervention was developed to modify impulsive behaviors and increase understanding about how one's actions impact other people (Ross & Fabiano, 1985). Wilkinson (2005) studied the effectiveness of the R&R program (measured as reconviction) among adult offenders participating in a community-based program in London. Due in part to small sample sizes for treatment and control group offenders and a lack of randomization, findings overall were mixed or difficult to interpret. Indeed, offenders who reported pro-social attitude changes were more likely to be reconvicted than offenders who reported more negative attitudes. Wilkinson (2005) concluded that the program "did not bring about significant reductions in offending" among a sample of high-risk repeat offenders who completed R&R compared with offenders in a control group not participating in the program (p. 81).

Other evidence about the effectiveness of R&R for reducing criminal behavior was gathered through meta-analyses. For example, Wilson et al., (2005) conducted a meta-analysis of three experimental evaluations and two high-quality quasi-experimental studies that reported the effectiveness of R&R for reducing recidivism rates. Although positive outcomes were found in all three experiments, in only two of the three were findings robust enough to reach statistical significance. Overall, Wilson et al., (2005) concluded "R&R is effective at reducing future criminal behavior" (p. 193); however, the small effect size and variation existing across study sites were causes for concern (see Robinson (1995), Porporino, Fabiano, & Robinson (1991), and Ross, Fabiano, & Ewles (1988) for specific study details).

A more recent meta-analysis was conducted by Tong and Farrington (2008) comparing evaluation findings from 19 different studies carried out mostly in the United Kingdom, United States, and Canada. Overall, findings revealed a 14 percent decrease in reconvictions among program participants compared with control groups. Tong and Farrington (2008) mentioned causes for concern were variations in methodological rigor across studies, a focus on reducing recidivism alone and not attitude change, and diminished effectiveness in more recent studies.

Findings shown here from three popular correctional programs indicated CBIs can reduce incidents of recidivism among offenders released from prison back into society. In general, CBIs are based on a theoretical framework which gradually focuses an offender's attention on developing better problem solving and decision-making skills, communicating with assertiveness and negotiation skills, controlled anger management, and critical reasoning (Brazão, da Motta, Rijo, do Céu Salvador, Pinto-Gouveia, & Ramos, 2015). Also, critical for the effectiveness of CBIs is an understanding that correctional environments are places for social learning. Indeed, correctional staff training should emphasize the role employees have for reinforcing positive behavior, including the understanding that inmates were always watching and learning from them (Antonio, Young, & Wingeard, 2009a; Antonio, Young, & Wingeard, 2009b; Dodge, 1993). What staff do and how they behavior outside of the classroom, will influence the overall outcomes of prison programming and community interventions. Overall, CBIs targeting a correctional population can be effective if they focus on factors that contribute to crime and programmatic materials that contribute to changing offender behavior (Latessa, Cullen, & Gendreau, 2002).

Many other programs have been created since Martinson's work to address the unique needs of prisoners. Before a prison program is selected and delivered to a correctional

population, it should be rigorously tested for effectiveness and potential impact on future behaviors. The present analysis performed such an evaluation of a recently developed CBI used with parolees after release from prison and returned to local communities throughout Pennsylvania.

Cognitive Life Skills Program and the National Curriculum & Training Institute

T4C, MRT, and R&R programs are popular offender initiatives that have been used by many correctional agencies throughout the world and have been studied empirically by numerous researchers. Another cognitive-behavioral program has become popular and is used widely for the purposes of decreasing criminal activities among paroled offenders. This intervention is referred to as the *Cognitive Life Skills* (CLS) program.

The CLS program was designed to be administered to medium to high-risk offenders in a group setting. Individual sessions address problems faced by offenders as they reenter the community. These problems, including addiction, employment, family, responsibility, and victimization, often serve as stumbling blocks impacting successful reintegration. Level 1 of the CLS program contains 11 sessions and approximately 16 hours of material, while Level 2 contains 22 sessions and provides 48 hours of content which allows for additional support and relationship building between the offender and facilitator (National Curriculum & Training Institute, 2012). The program facilitator and other appropriate staff will determine which program level is suitable for the offender based on individual needs. Overall, the program's learning objectives are to improve critical cognitive thinking skills, gain better self-control, develop an understanding for how attitude affects behavior, and facilitate successful reintegration into society.

The CLS program is one of 17 offense-specific CBI programs designed by NCTI to target the needs of adult offenders. Each program is equipped with 1) a user's manual that guides the offender through dozens of curriculum components and 2) an instructor's manual that facilitates delivery of the program, including organized homework assignments based on lesson content, open-ended questions to direct curriculum-specific conversation, and graduated role-play scenarios. The CLS program was designed specifically to improve an offender's anti-social attitudes and poor decision-making and problem solving skills, and address individual needs for substance abuse treatment and job readiness training (National Curriculum & Training Institute, 2011).

Pennsylvania's Board of Probation & Parole

One criminal justice agency that utilizes the CLS program is Pennsylvania's Board of Probation & Parole (PBPP). It is the mission of PBPP to protect the safety of the public, while ensuring the custody, control, and treatment of offenders under its supervision. To achieve its mission, PBPP has devoted considerable resources to facilitating offender transition from prison back into the community. "This commitment to community reentry has involved addressing offenders' individual treatment needs through rehabilitative programming, assessing offenders' high-risk factors for reoffending, and educating staff about the benefits of programming on offenders' well-being and for reducing recidivism rates" (Antonio & Klunk, 2014, p. 43).

Over the past decade, PBPP has made significant progress related to facilitating offender reintegration back into society. Most recently, research findings from an 18-month, multi-site evaluation of an offender reentry program revealed measurable improvements among offenders' attitudes and behaviors for illicit substance use, cognitive distortions, and anti-social

associations, for which offenders reported avoiding negative people and circumstances; improving relationships with family, friends, and co-workers; attending school; retaining employment; and increasing income (Zortman, Powers, Hiester, Klunk, & Antonio, 2016). In 2009, PBPP created Assessment Sanctioning and Community Resource Agent (ASCRA) positions for the purpose of ensuring "…that the board policy of progressive, corrective and rehabilitative management of parolees is achieved consistently across all parole districts, with the objective of managing parolees in the communities as long as they can be safely and effectively supervised" (Pennsylvania Board of Probation & Parole, 2009, p. 1). The ASCRAs are chiefly responsible for delivering a wide variety of CBIs to offenders who are under supervision by PBPP. ASCRAs started delivering the CLS program in 2010.

As of 2016, the CLS program was facilitated by ASCRAs located in parole offices throughout the Commonwealth of Pennsylvania. Each facilitator received training and was certified by the NCTI to deliver the program. When possible, CLS program sessions were facilitated in a private and secure room within the PBPP district offices or sub-offices. Each session was designed to last one to two hours in duration and the facilitators were trained to deliver the program in two to three sessions per week. The number of participants in each class could vary from approximately five to twenty.

Offender placement in the CLS program may occur in two ways: 1) assignment by the supervising parole agent or 2) ordered by a Board Member as a condition of release from prison. Typically, offenders are considered appropriate for inclusion in a CLS program if they meet certain criteria: medium or high risk for reoffending categories (see Pennsylvania Board of Probation & Parole, 2010); a Board Member determines programming is needed upon reentering

the community; the maximum supervision date does not occur prior to the anticipated program completion date; and/or the supervising agent determines programming is needed.

The Present Analysis

While the CLS program has been delivered to offender populations in numerous jurisdictions through the country, no rigorous, empirically-based evaluation has been performed to determine the effectiveness of the program related to treatment outcomes. The present analysis intended to remedy this by examining data gathered from PBPP about offenders who completed NCTI's CLS program. This analysis was performed for several purposes. First, the main purpose was to determine program outcomes, including recidivism measured as re-incarceration for committing new crimes and/or for violating technical conditions of parole, among offenders who completed CLS compared with offenders who did not receive the program. Another purpose for this analysis was to provide PBPP with outcome findings that would guide future development and use of cognitive-behavioral programming for facilitating offender reentry into the community. Finally, it was hoped that findings from this analysis would be utilized by criminal justice professionals in other jurisdictions and would facilitate programmatic changes related to offender programming and rehabilitation efforts.

Method

Data

The data used in this analysis was provided by PBPP, Statistical Reporting and Evidence-Based Program Evaluation Office in the spring of 2015. Altogether, records were obtained for

87,577 offenders who were released to parole supervision between January 2010 and May 2015. The data file contained variables critical for a program evaluation of this scope and nature including: offender characteristics, date of release to PBPP supervision, CLS program enrollment/completion dates, risk assessment score, and re-incarceration information resulting from a new crime and/or a violation of technical conditions of parole. The compressed data file was received from PBPP on a password-protected DVD. A password was sent under separate cover to ensure protection of confidential information related to offender records. All analyses were performed using the statistical software R and SPSS on secured databases to safeguard sensitive information. The purpose of the present analysis was to examine the overall effectiveness of the CLS program by comparing incidents of recidivism, measured as re-incarceration, between offenders completing CLS with a matched group of offenders not receiving it.

Matching Variables

Several important matching variables were examined in this analysis. These included offender gender (female/male), age (years), race (White, Black, or Hispanic), location of release (district office), offense category (e.g., homicide, sexual offense, robbery, etc.), and history of violence (no/yes). Also, information was provided about the most recent risk assessment score for each offender, as determined by the Level of Service Inventory-Revised (LSI-R) instrument. LSI-R risk level was determined by the last assessment administered to the offender prior to his or her release to PBPP supervision.

The LSI-R is an assessment tool that provides a score related to an offender's risk for reoffending. It is recognized as a reliable and valid instrument for assessing risk factors related to

community recidivism and is used to inform parole decision-making (Andrews & Bonta, 2003b; Manchak, Skeem, & Douglas, 2008). Assessment findings should be used to identify offenders with the highest priority for treatment, regardless of their programmatic needs or specific problem areas. The LSI-R is diverse and can be used with adult male or female offenders regardless of the offense category, including those incarcerated in prison and while under supervision in the community. The assessment tool contained 54 items and was comprised of 10 domains: criminal history, education/employment, financial, family/marital, accommodation, leisure/recreation, companions, alcohol/drug problems, emotional/personal, and attitudes/orientation.

Scores on the LSI-R range from 0, indicating no risk for reoffending, to 54 indicating significant tendencies for future criminal behavior. PBPP utilizes three cutoff points to establish offender risk and need for future programming. For this analysis, offenders deemed high-risk for reoffending scored 28-54, medium-risk offenders scored 18-27, and low-risk offenders scored 17 or lower. PBPP policy indicates programming should be reserved for offenders with medium or high LSI-R scores only (see Pennsylvania Board of Probation & Parole, 2010).

Matching is a common tool used in observational studies to minimize the possibility of confounding variables. Since the treatment level is the primary explanatory variable of interest in this analysis, it is important that offenders who received the treatment be matched with offenders who were equally likely to receive the treatment, but did not. In recent years, propensity score matching has been a widely used technique among evaluation researchers to create and assess for statistical equivalence among known characteristics of participants in treatment conditions (Peters, Hochstetler, DeLisi, & Kuo, 2015). It is important that offenders with varying characteristics are randomly assigned into groups, where one group gets an intervention program

(cases) and the other does not (controls), and that both groups are based on similar underlying probabilities of being enrolled into the treatment itself. Here logistic regression was used to estimate the probability that a parolee would be enrolled in the program given other factors.

In addition to the matching factors shown above, the year when an offender successfully completed the CLS program was matched to the year of release to parole supervision for control group offenders. Let Y = 1 indicate an offender who completed the program (treatment group) and Y = 0 indicate an offender who did not receive the treatment (control group). Define π_i = P(Yi = 1|x), where x is a vector of values for the grouping factors. The final generalized linear model was defined as:

$$v_i = \ln\ (\pi_i \mid 1-\pi_i) = \beta_0 + \beta_1 \text{location}_i + \beta_2 \text{risk}_i + \beta_3 \text{gender}_i + \beta_4 \text{age}_i + \beta_5 \text{ethnicity}_i + \beta_6 \text{offense category}_i + \beta_7 \text{history of violence}_i + \beta_8 \text{release year}_i$$

Figure 1 illustrates that the majority of offenders under parole supervision were unlikely to have been enrolled in the CLS program. Findings revealed the average propensity score of the control group offenders was .035 compared to .081 for the treatment group. Overall, there still was significant overlap between the density curves which made possible matching those offenders who received the program (cases) with offender who did not (controls) based on similar underlying probabilities of getting the treatment.

Figure 1: Propensity Score Overlap

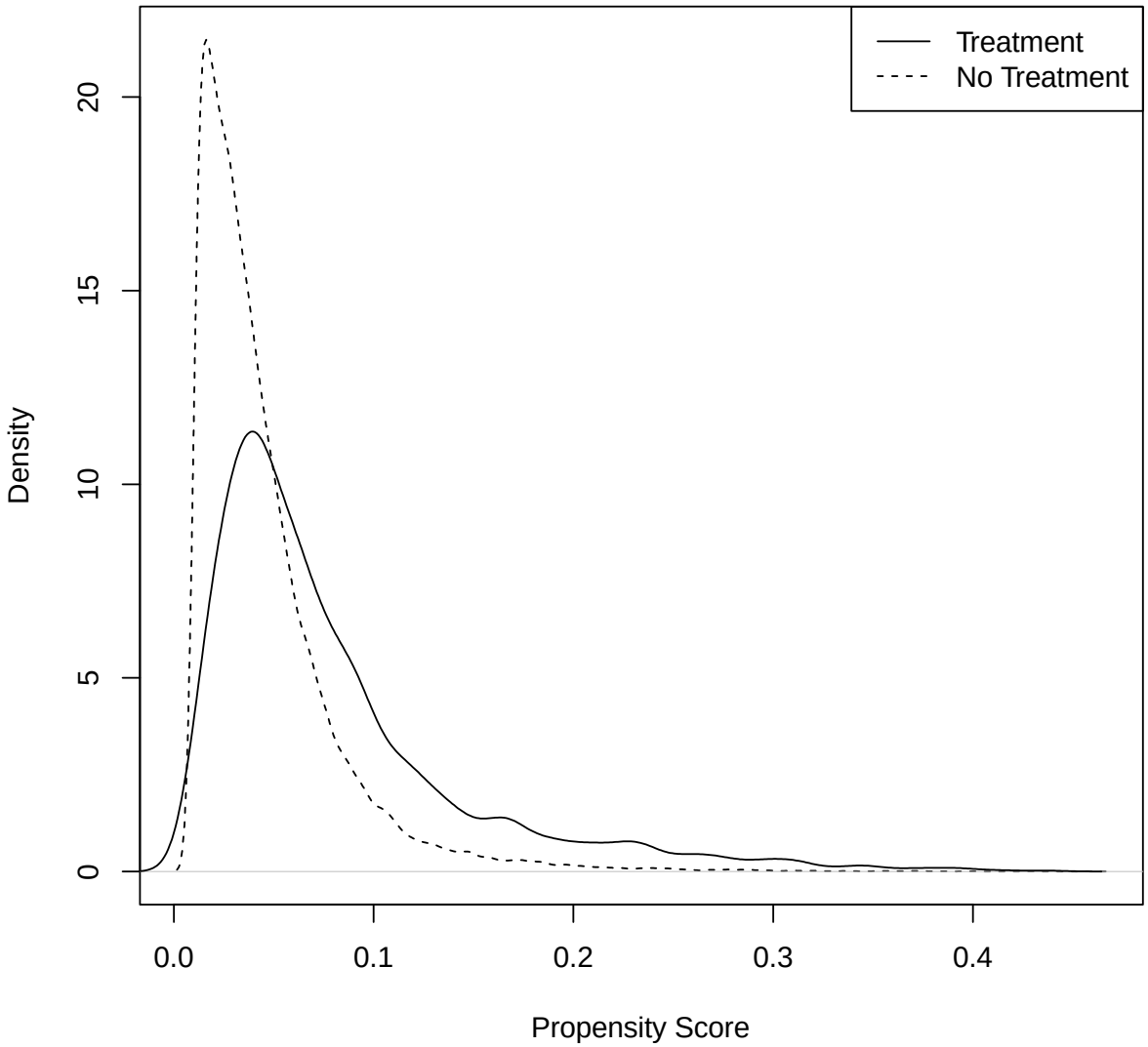


Table 1 displays the results from the logistic regression model used for propensity score matching for the full sample of offenders considered in this analysis (4,230). As can be seen, ethnicity, location, LSI-R score, sentenced for a sexual offense, and the year released to parole supervision were significant predictors for whether the offender received the CLS program. A negative value in the second column (β) or value less than 1 in the fourth column ($Exp(\beta)$)

predicted that the offender was less likely to receive the program, holding the other covariates constant. It can be seen that location and year of release had the largest effects on the odds of the offender receiving the program. The large negative values found for the terms related to the intercept, Williamsport (location), and Mercer (location) occurred because only about one percent of all offenders received the program in 2010 (baseline for year of release), .05% received the program from Mercer overall, and 0.11% received the program from Williamsport overall. Therefore, if released to any location during 2010 or being released from Mercer or Williamsport at any time between 2010-2015, it was highly unlikely the offender would receive the treatment program.

Table 1: Logistic Regression Model

Variables	Full Sample (N=4,230) *β*	*SE*	*Exp(β)*
Intercept	-6.61	.241	0.001***
Offender Characteristics			
Gender (male)	-0.07	.079	0.929
Age	0.003	.002	1.003
Hispanic	-0.25	.084	0.781**
White	-0.11	.056	0.897
Location			
Pittsburgh	0.23	.085	1.255**
Harrisburg	0.80	.072	2.237***
Scranton	0.60	.104	1.849***
Williamsport	-2.98	.711	0.051***
Erie	1.84	.085	6.424***
Allentown	0.41	.084	1.514***
Mercer	-3.48	.710	0.031***
Altoona	1.04	.111	2.866***
Chester	0.19	.106	1.200
Offense Categories			
Sex Offense	-0.62	.228	0.520**
Robbery	-0.05	.185	0.966
Aggravated Assault	-0.01	.191	0.993
Burglary	-0.16	.191	0.859
Arson	0.12	.340	1.132
Other Violent Offense	-0.31	.747	0.729
Firearms	0.07	.197	1.070
Non-violent Assaults	-0.45	.263	0.635
Drugs	0.10	.176	1.114
DUI	0.05	.193	1.049
Theft and Fraud	-0.16	.184	0.859
Other Non-violent	-0.28	.181	0.752
Other			
Risk score	0.06	.003	1.058***
Violent behavior	0.09	.054	1.091
Year			
2011	0.92	.122	2.505***
2012	1.58	.113	4.864***
2013	2.06	.111	7.831***
2014	1.79	.113	5.975***
2015	1.65	.136	5.210***
Model diagnostics			
R^2	0.112		

** p <.01, ***p<.001

Dependent Variable

Because of the large number of records contained in the database provided by PBPP, the true effectiveness of the CLS program could be assessed by identifying a recidivism event occurring after the completion of the program. Recidivism was measured in this analysis as re-incarceration resulting from commission of new crimes (Criminal Parole Violation=CPV) and/or for violating technical conditions of parole (Technical Parole Violation=TPV). For offenders completing the CLS program, time to recidivate (days) was determined by subtracting the date they completed the program from the date of the first recidivism event. For offenders not receiving the CLS program, time to recidivate was determined by subtracting the date they were released to parole supervision from the date of the first recidivism event.

Analytical Procedures

A series of analyses were performed to assess the effectiveness of the CLS program for reducing future incidents of recidivism. First, summary statistics were obtained for the main matching variables and findings compared between the treatment and control group offenders. Next, the effect of the CLS program on incidents of recidivism was estimated using a Cox Proportional Hazard Model. This technique estimated the instantaneous probability of recidivating, given the offender had yet to recidivate at a given point in time. The final model estimate was defined as:

$$hi(t) = \lim_{\Delta t \to 0} \frac{P(t < Ti \leq t + \Delta t | Ti > t)}{\Delta t}$$

where Ti indicates the time at which offender *i* recidivated. The final Cox Hazard Model used to estimate risk was :

$$\ln(h_i(t)) = \beta_1(t) + \beta x_i$$

where x_i is the vector of covariates for offender *i*. This implies that the probability of offender *i* recidivating presently is $e^{\beta_1(t) + \beta x_i}$.

Missing Data

The data file contained records for approximately 700 offenders who started the program, but never successfully completed it. Program facilitators would determine successful completion based on the number of sessions attended and demonstrated mastery of program materials. Often offender records did not specify the reason(s) for not completing the CLS program other than being labeled as "unsuccessful". Some records did indicate unsuccessful completion was based on the offender's misbehavior, including an absconding event or a violation of a condition of parole. Also, unsuccessful completion resulted from non-criminal factors, including medical discharges, obtaining employment, and transfer to other district offices or out of state. Because not completing the program could be justified for either criminal or non-criminal behavior and because an exact number of sessions attended could not be confirmed, only offenders whose record indicated successful completion of CLS were included in this analysis.

Finally, offender records were removed from the final analysis due to discrepancies in the dates for participating in the CLS program. For example, an offender was removed if the program completion date was missing or it was recorded as occurring before the program start day, and if the date of participation in the program was recorded as occurring before the offender's release to parole supervision. Control group offenders were removed from the final analysis as well due to date discrepancies, including finding a date of recidivism occurred prior to release to parole supervision. Altogether, this eliminated 1,686 offenders from the analysis.

Results

Offender Characteristics

Table 2 reveals findings about offender characteristics used to create the matched groups. Overall, a total of 4,230 offenders' supervision histories were examined in this this analysis. When the full sample of offenders was considered, it was found that the majority were male (90.4%), young (median 34), and similarly racially represented between White and Black (47.5% and 43.7%, respectively). Over 40 percent of the offenders were released to parole supervision in two of the 10 district offices located throughout Pennsylvania, namely, Philadelphia (21.6%) and Harrisburg (22.0%). The majority of offenders scored either in the medium (35.6%) or high (57.4%) level on their latest LSI-R risk assessment and one-third (33.4%) served a sentence for a drug related crime prior to release to parole supervision.

Table 2 also shows findings about offender characteristics when separated into the matched treatment and control groups (2,115 and 2,115 respectively). The median number of days for offenders to complete the program was 49 (not shown). All treatment group offenders included in this analysis completed the CLS program and contained complete records for all key variables understudy. As can be seen in the table, a chi-square test statistic revealed the two groups were statistically equivalent on many of the matching criteria. Small percentage differences were found between groups for risk level and previous incarceration for a drug related crime.

Table 2: Propensity Matching Controls

Variable		Group		
	Full Sample	Treatment	Control	
	N=4,230	N=2,115	N=2,115	Sign.
Gender (% male)	90.4	90.4	90.3	
Age (median years)	34	34	34	
Race (%)				
White	47.5	47.5	47.5	
Black	43.7	43.3	44.1	
Hispanic	8.8	9.2	8.5	
Location of Release				
Philadelphia	21.6	20.3	22.8	
Pittsburgh	11.2	11.7	10.7	
Harrisburg	22.0	21.9	22.1	
Scranton	6.6	6.7	6.5	
Williamsport	.07	.09	.04	
Erie	14.8	14.7	15.0	
Allentown	11.7	12.3	11.0	
Mercer	.12	.09	.14	
Altoona	6.4	6.4	6.4	
Chester	5.5	5.8	5.2	
Risk Score (%)				
Low	7.1	6.5	7.7	
Medium	35.6	33.9	37.3	*
High	57.4	59.7	55.0	**
Current Offense (%)				
Criminal Homicide	1.6	1.7	1.5	
Sexual Offense	2.5	2.2	2.8	
Robbery	10.2	10.0	10.4	
Aggravated Assault	6.5	6.7	6.3	
Burglary	7.4	7.2	7.5	
Arson	.54	.61	.47	
Other Violent Offenses	.09	.09	.09	
Firearms	5.0	5.2	4.8	
Non-violent Assaults	1.2	1.3	1.2	
Drugs	33.4	31.9	34.8	*
DUI	6.7	6.9	6.5	
Theft and Fraud	11.3	11.8	10.7	
Other Non-violent	13.6	14.3	13.0	
Violent History (% yes)	31.4	32.0	30.8	
Year				
2010	5.0	4.5	5.4	
2011	11.2	11.3	11.0	
2012	20.9	20.9	20.9	
2013	32.5	32.2	32.8	
2014	24.6	24.6	24.5	
2015	5.9	6.4	5.4	

*$p \leq .05$, ** $p < .01$

Recidivism

Table 3 reports findings about the main dependent variable, namely, a recidivism event that occurred while under parole supervision. Recidivism was measured in this analysis as a violation of a condition for parole that resulted in re-incarceration. As can be seen in the table, 23.3 percent of the offenders, those who completed the CLS program, engaged in violations of conditions for parole and/or some type of criminal behavior while under supervision by PBPP that resulted in re-incarcerated. A greater percentage of the matched control group offenders, those who did not receive the CLS program, were re-incarcerated because of similar misbehavior (27.3%). This four percentage point difference in recidivism between the two groups was statistically significant beyond the .01 probability level. The type of violation(s) committed (CPV, TPV, or both) did not differ between the matched groups.

Also shown in Table 3 are recidivism findings among offenders who were found to be a high risk for reoffending based on their latest LSI-R assessment. Overall, 23.9 percent of high risk offenders in the treatment group recidivated compared to 31.7 percent of offenders in the matched control group. This is approximately an eight percentage point difference and is statistically significant beyond the .001 probability level. Findings also showed differences for the type of violation committed by group of offenders. Specifically, offenders who completed the CLS program were re-incarcerated for CPVs, while control group offenders were re-incarcerated for TPVs.

Table 3: Comparison of Recidivism Event between Treatment and Control Group Offenders

Variable	Full Sample Treatment N=2115	Control N=2115	Sign.	High Risk Offenders Only Treatment N=1262	Control N=1164	Sign.
Recidivate (% re-incarcerated)	23.3	27.3	**	23.9	31.7	***
CPV	18.3	14.4		17.2	11.4	*
TPV	79.5	83.9		81.1	87.0	*
Both CPV and TPV	2.2	1.7		1.7	1.6	

*p ≤ .05, ** p <.01, ***p < .001

Hazard Function Model

Table 4 reveals findings related to a survival analysis using a Cox proportional hazard model. As can be seen in the table, completing the CLS program was a highly significant predictor for not recidivating with a p-value less than .001. The exponentiated coefficients, *Exp(B)*, are the multiplicative effect on the hazard function. In other words, holding the other covariates constant, offenders completing the CLS program reduced the daily probability of re-incarceration by a factor of 0.76, on average. This translates to a 24 percent reduction in recidivism. Indeed, the 95% confidence interval for the parameter is (0.67, 0.85), indicating at worst a 15 percent reduction and at best a 33 percent reduction in recidivism compared to matched group offenders not receiving the program. Also, gender, age and the criminal history subdomain on the LSI-R were significant predictors for the probability of a recidivating event (p-value < .001). Here, males were much more likely to recidivate and for every additional year at risk, the average daily probability of recidivating decreased by 3 percent, holding the other covariates constant.

Table 4 also includes parameter estimates and standard errors for the high-risk offenders exclusively. For these offenders, the same covariates were significant in the model (p-value <

.001), but the treatment effect is much larger. Here, an *Exp(B)* value of 0.69, indicates the daily probability of re-incarceration for the CLS program completers decreased by 31 percent on average, holding all other covariates constant. The 95% confidence interval is (0.59, 0.81). In both the full sample and the high-risk offenders there is strong evidence that receiving the treatment program significantly reduces the probability of the offender recidivating by a substantial margin.

Table 4: Cox Proportional Hazard Model

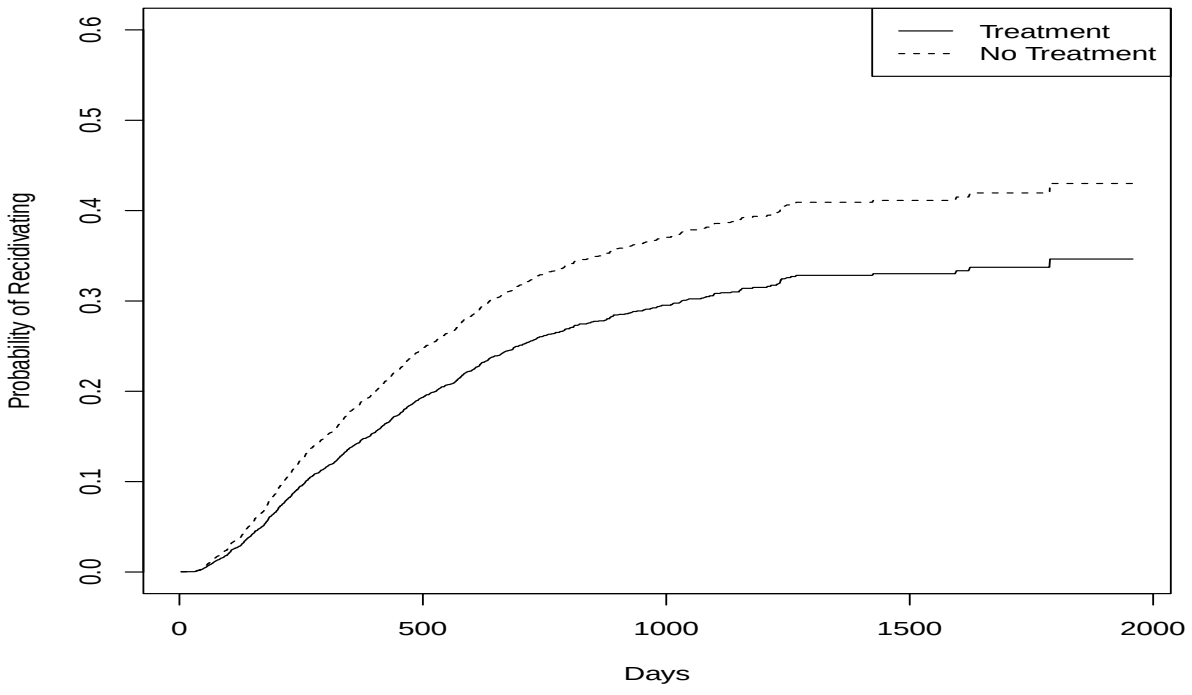
	Full Sample				High Risk Offenders Only			
Variables	*B*	*SE*	*Exp(B)*	95% CI	*B*	*SE*	*Exp(B)*	95% CI
Group								
Treatment	-0.28	.061	0.76***	(0.67,0.85)	-0.37	.079	0.69***	(0.59,0.81)
Offender Characteristics								
Gender (male)	0.50	.134	1.65***	(1.27,2.14)	0.55	.166	1.73***	(1.25,2.39)
Age	-0.03	.003	0.97***	(0.96,0.97)	-0.03	.004	0.97***	(0.96,0.98)
Risk Assessment Subdomain								
Criminal History	0.13	.019	1.14***	(1.10,1.19)	0.09	.029	1.10***	(1.04,1.16)
Educational/employment	-0.004	.016	0.99	(0.96,1.03)	0.00	.026	1.00	(0.95,1.05)
Financial	-0.04	.047	0.97	(0.88,1.06)	-0.06	.062	0.94	(0.83,1.06)
Family/marital	0.04	.026	1.05	(1.00,1.10)	0.05	.032	1.05	(0.98,1.12)
Accommodations	0.05	.039	1.05	(0.97,1.13)	0.04	.047	1.05	(0.95,1.14)
Leisure/recreation	0.004	.037	1.00	(0.93,1.08)	-0.06	.050	0.95	(0.86,1.04)
Anti-social Companions	0.05	.030	1.05	(0.99,1.12)	0.01	.043	1.01	(0.93,1.10)
Alcohol/drug Problems	0.02	.014	1.02	(0.99,1.05)	0.02	.021	1.02	(0.98,1.07)
Emotional/personal	0.04	.021	1.04	(0.99,1.08)	0.05	.026	1.05	(0.99,1.11)
Attitudes/orientation	-0.02	.026	0.98	(0.93,1.03)	-0.03	.030	0.97	(0.92,1.03)
Offense Category								
Violent Behavior	0.11	.065	1.12	(0.98,1.27)	0.07	0.080	1.068	(0.91,1.25)
Model Diagnostics								
Model χ^2	193.2***				105.3***			
R^2	0.088				0.085			

*** p <.001

Figure 2 illustrates the distribution of the cumulative hazard function over the number of days in the study for the sample. Because the purpose of the study was to assess the impact of the CLS program on recidivism, the modeled probability of recidivating for the cases and the controls was examined, holding the other covariates at their respective means (or baseline value). As can be seen from the figure, after a brief number of days, the dashed line (controls) was

consistently above the solid line (cases). This indicates that holding the other covariates constant, the probability of daily recidivism was always higher on average for those who did not receive the CLS program than it was for offenders who did complete it, regardless of the number of days they have been under parole supervision. In fact, as the number of days under parole supervision increased, the gap between the two groups tended to widen until reaching some asymptotic effect. This is consistent with a cursory examination of the full sample, where 14% of the cases had recidivated within a year versus 16% of the controls. After two years, the gap widens to 21% and 25%, respectively. The gap then stays consistent with about 23% of the cases and 27% of the controls recidivating by the end of the third year.

Figure 2: Hazard Function for At-Risk Time to Recidivate



Discussion

The present analysis was the first attempt to evaluate the effectiveness of the NCTI's CLS program for reducing recidivism among parolees under the supervision of PBPP. Using techniques associated with propensity score matching, a group of offenders who completed the CLS program was compared with a statistically equivalent group of offenders who did not receive the program. The main outcome of interest, recidivism, was measured as a violation of conditions for parole supervision that resulted in re-incarceration.

Outcome findings showed a positive impact among offenders who completed the CLS program compared to offenders not receiving it. First, the overall percentage of recidivism for CLS program completers was lower compared to a matched group of offenders not participating in the program. The effect was uncovered in matched comparisons for the full sample of offenders (4, 230) and when the analysis was conducted for the high-risk offenders only (2,426). For the full sample of offenders, there was a four percentage point reduction in recidivism between the treatment and control group. Comparison among the high risk offenders showed the most reduction in recidivism, which was approximately eight percentage points lower for the treatment group offenders. This finding is significant from a policy perspective.

Since Martinson, evaluation researchers have focused more heavily on assessing the individual characteristics or traits of the offender to predict reductions in future recidivism. Prison programs should be reserved for offenders who are medium or high risk for reoffending, as these offenders may experience the largest reductions in recidivism resulting from intensive correctional treatment (Andrews & Bonta, 2003a). On the other hand, low risk offenders should not be targeted for prison programming and enrolling them into such interventions may place

them at greater risk for future criminal behavior (Andrews, Bonta, & Hoge, 1990; Bonta, Wallace-Capretta, & Rooney, 2000). Findings uncovered in this analysis reinforce the idea that correctional programming should be reserved for those offenders who are at the highest risk for reoffending.

It was troubling to uncover that a greater percent of offenders completing the CLS program (compared to control group offenders) were returned to prison for misbehaviors related to new crimes (CPVs). This finding may suggest the program improved daily decision-making and problem solving skills because offenders were not returned as frequently for technical violations, however this finding indicates offenders still engaged in new criminal behavior that resulted in a return to prison. The overarching findings for reductions in recidivism, however, suggest that the CLS curriculum did address critical issues related to criminogenic needs that current evidenced-based practices indicate are associated with criminal behavior (Andrews & Bonta, 2003a; Andrews & Bonta, 2010). Also, these findings reinforce the commitment that PBPP has to CBI strategies and shows clear evidence that program facilitators and staff play a critical role for delivering effective programming to its supervised offender population.

Effectiveness of the CLS program also was demonstrated through findings from a survival analysis using a Cox proportional hazard model. General findings, consistent with previous research, showed males and offenders with a criminal history were more likely to recidivate, while older offenders were not. The most important findings from the hazard function analysis revealed, on average, a large 24 percent reduction in recidivism for CLS program completers (full sample) and, on average, a 31 percent reduction among high risk offenders exclusively, while holding other covariates constant. Also, findings illustrated that daily

recidivism was always higher of offenders who did not receive the program compared to those who successful completed it.

Limitations

There are several limitations that could have impacted the results in this evaluation. The first concern was that the main dependent variable measured offender recidivism only as re-incarceration due to commission of new crimes or violations of technical conditions for parole. Some research indicated studying recidivism as arrest for criminal behavior provides a better measure of program effectiveness. While this may have been beneficial, arrest records were not available in data procured from PBPP and therefore necessitated a return to prison as the main measure for recidivism exclusively. Second, removing offenders who did not successfully complete the CLS program may have introduced selection bias into the findings. However, because the reason(s) for not completing the program varied between criminal and non-criminal circumstances and often times reasons for removal were unclear, as described above, it was believed focusing on program completers only would provide the best comparison and assessment for the true effectiveness of the CLS program. Finally, it would have been insightful to control for offenders' prior programming and treatment experiences. Again, because of difficulties obtaining records for offenders' treatment history, this issue was not be examined in the present analysis.

Conclusion

Given these few drawbacks, the present analysis was characterized by many advantages that bolstered the statistically significant relationships uncovered here. First, the number of offenders included in this single evaluation far exceeded published findings from most other

single evaluation studies in print. This should instill confidence for findings shown here. Second, in many cases, the follow-up period for recidivism exceeded one year, which appears to be the standard for many other evaluations of this nature. Being able to conduct a two- or three-year follow-up for at-risk time for recidivism reveals the long-term effects of the program and gives direction for needed areas of improvement and change. This design allowed for a more comprehensive understanding about the true impact of the CLS program on future criminal behavior.

In conclusion, findings shown here marked the first attempt for conducting an empirically rigorous evaluation of NCTI's CLS program. However, results generated from any one analysis cannot serve as the defining outcome for the effectiveness of a program. Indeed, replication of these evaluation findings elsewhere will speak to the generalizability of the program material for reducing recidivism among offenders in other jurisdictions. Therefore, additional analyses and measures of recidivism should be performed in other criminal justice agencies for offenders who vary in age, gender, size, and crime type.

"This article does not contain any studies with human participants or animals performed by any of the authors."

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